

# Portafolio

# UPEN

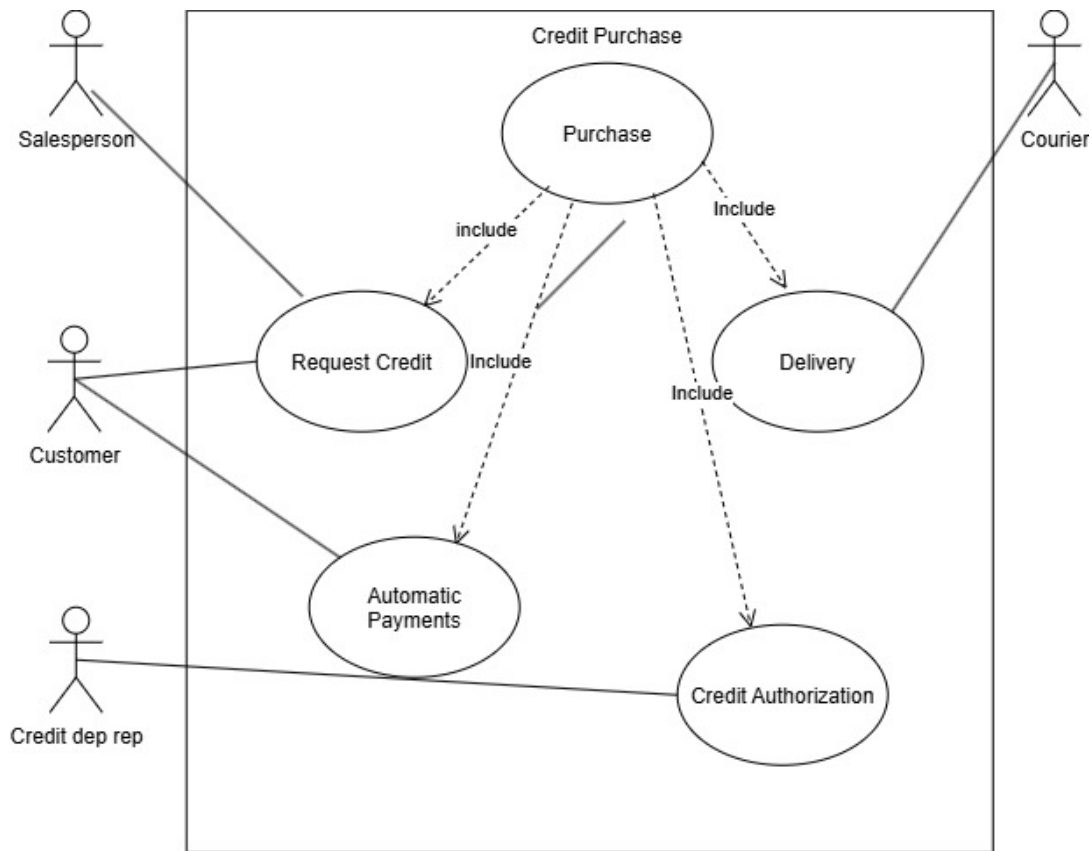
## Unidad 1

```
1 class Galleta:
2     def __init__(self, sabor, tamaño, color):
3         self.sabor = sabor
4         self.tamaño = tamaño
5         self.color = color
6
7     def comer(self):
8         print("Me estoy comiendo la galleta")
9
10    def describir(self):
11        print(f"Es una galleta de {self.sabor}, tamaño {self.tamaño} y color {self.color}")
```

A class is like a mold used to create objects. For example, we can create a class called `Cookie` and define its characteristics—such as flavor, size, and color—within it.

Methods are the actions that the object can perform. In the case of a cookie, it might have methods like `eat()` or `describe()`. Simply put, the class defines what the object is like, while the methods determine what it can do.

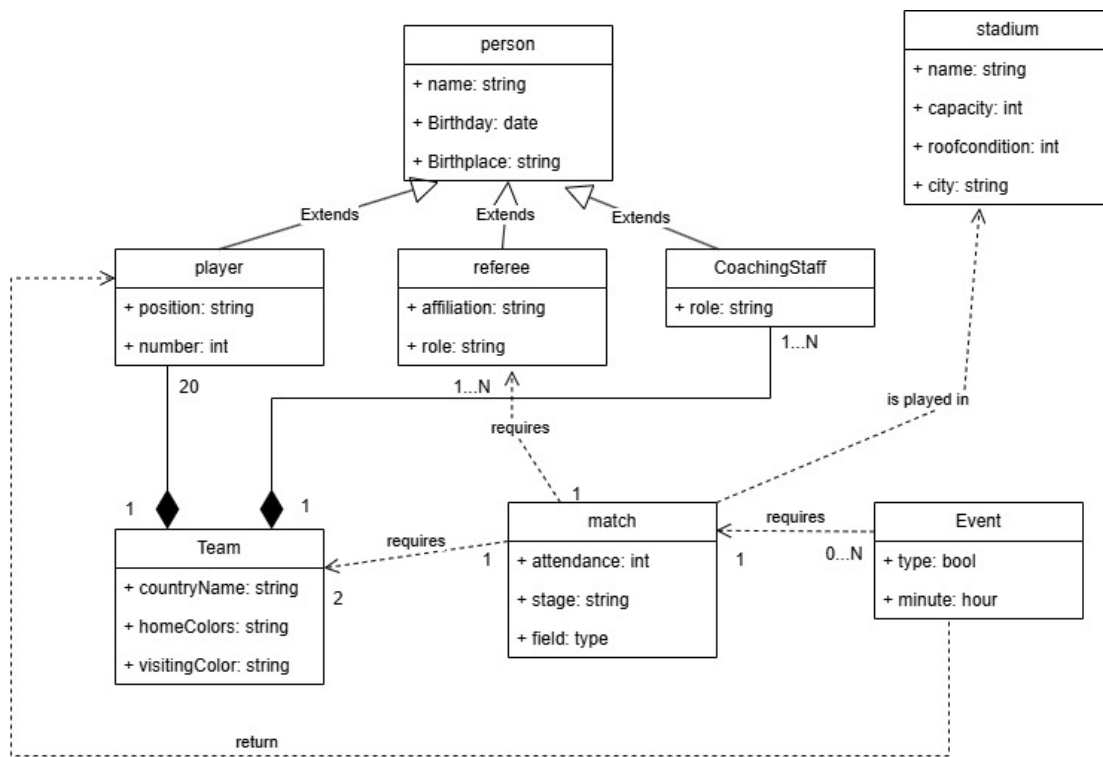
## Credit diagram



## Improvement

The diagram can be made clearer by using the UML notation <> instead of simply writing "include." In addition, the question discussed in class can be addressed by establishing a connection between the credit application and the credit authorization, since both processes are part of the same area of responsibility. Finally, the actor should be labeled "Credit Department Representative" because this name more accurately describes their role in the system.

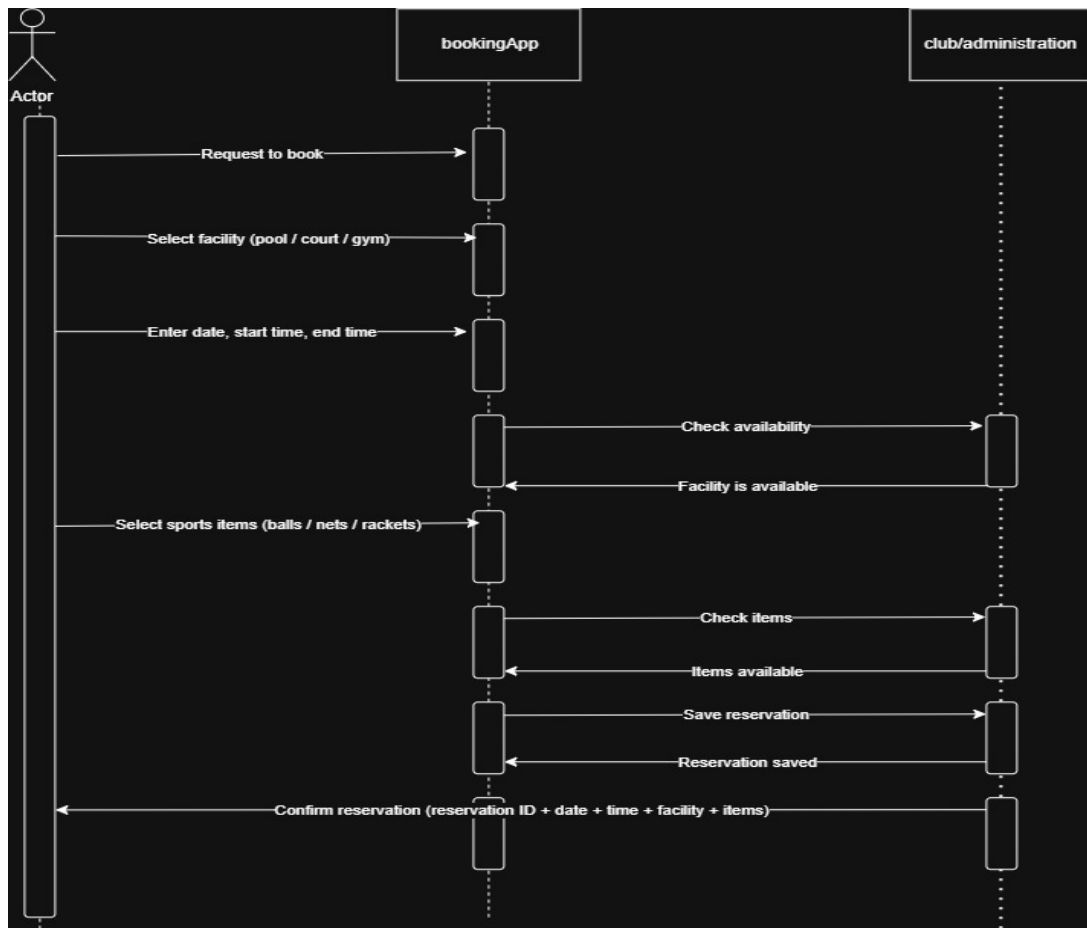
## Soccer match diagram



## Improvement

There are some problems with the arrows in the diagram. Player, Referee, and Coach come from the Person class, so the arrow should point to Person and use a white triangle. The general idea is correct, but there are some things that should be changed. The type: bool part is one of the errors. Also, the relationship between Match and Stadium should be a dotted line, and the relationship between Team and Player should be a composition.

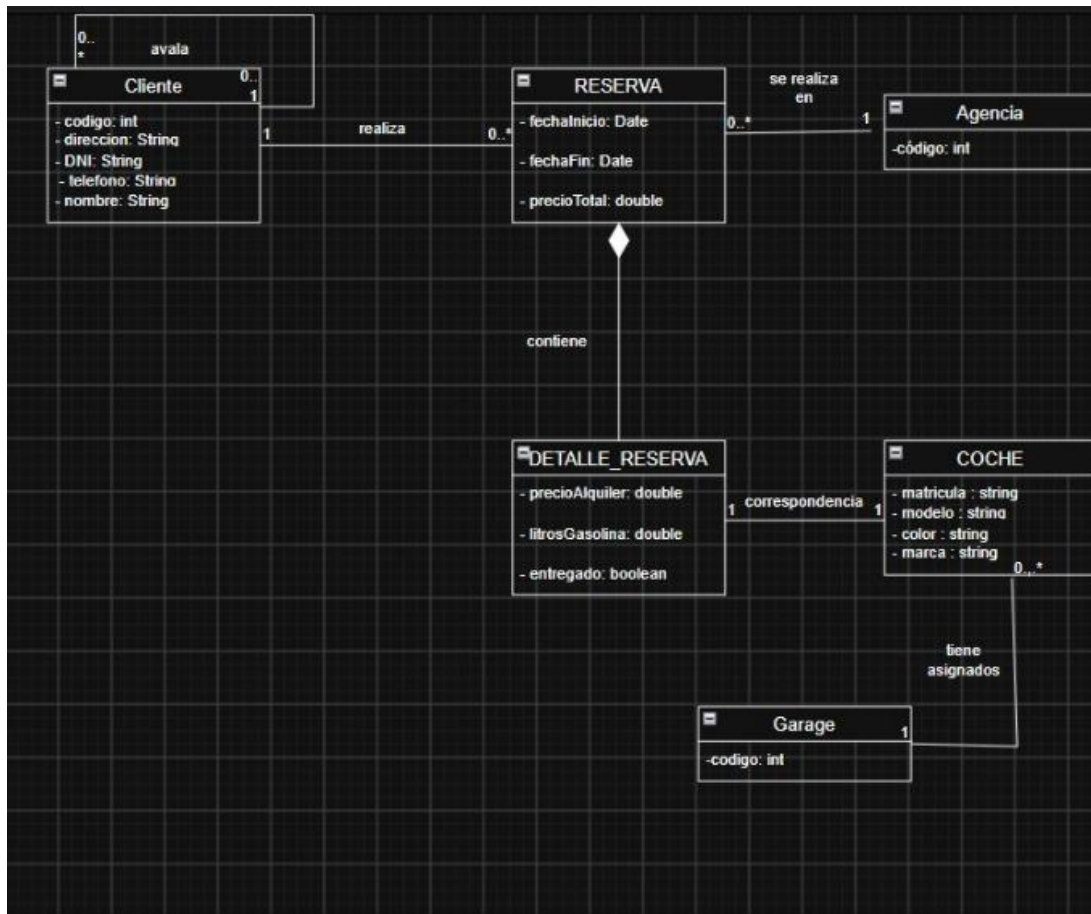
## Court reservation



## Improvement

This diagram is better than the previous one, but it still has some things that can be improved. For example, we can change “actor” to “client” because it is easier to understand. The names of the applications can also be changed to “BOOKING APP” and “CLUB ADMINISTRATION.” At the end, the message can say “RESERVATION CONFIRMED.” We can also add an alt condition to show what happens when the reservation is not successful.

## Car reservation



## Improvement

The Agency class is missing some information because it only has a code. We could add things like the name and phone number. The ID could also have more information if it is made from different parts. Besides that, I don't see any other important errors in the diagram.

## **Conclusion**

In conclusion, these diagrams help us understand how different systems work. The class diagram helped us see the attributes, data types, and relationships between the classes. The use case diagram showed the actions that the actors can do in an online shopping system. Finally, the sequence diagram showed the order in which the messages happen.